Overview

Who Are We?

We are students of Cohort 6 in the MIT Emerging Talent program. Through this program, we develop our understanding of computer science and data science concepts, as well as build our coding expertise. Currently, we are engaged in the initial phase of the Experiential Learning Opportunity (ELO) stage. This phase includes a Collaborative Data Science Project (CDSP), where we examine the complete data science lifecycle. Throughout this project, we apply our knowledge across key milestones, from problem identification to communicating results.

What We Do

As students from war-affected regions, including Palestine, Ukraine, Yemen, and others, we are particularly motivated to support fellow students impacted by war and conflict. Our objective is to examine the effects of active conflict and war on students' academic performance, especially following the shift to online learning due to the closure or destruction of educational institutions. In this study, we are primarily focusing on Palestinian college students, with particular emphasis on the ongoing war in Gaza.

Problem Statement

In regions impacted by war, the educational system is often among the first societal structures to deteriorate. Students who previously demonstrated high levels of academic achievement and consistent attendance frequently experience significant declines in both areas once conflict begins. Contributing factors include violence, displacement, psychological trauma, and the breakdown of essential infrastructure such as internet connectivity and school availability. This project seeks to examine how armed conflict affects students' academic performance and attendance over time. By comparing academic records from before and during the conflict, we aim to measure the extent of disruption and identify patterns that can guide targeted interventions.

Research Question

How does war disruption, such as violence, displacement, and infrastructure breakdown, impact students' academic performance, engagement, and attendance in online learning for college students in Gaza?

Data Access and Current Limitations

We have analyzed war events data to measure the impact of war on students and the education system in Gaza. Some of our insights:

- We found that over 500 schools have been destroyed or damaged. That's over 80% of all schools in the region.
- **99.3%** of the damaged infrastructure was **residential**. This means students' houses were destroyed.
- **Excluding the residential damage**, **32.5%** of the infrastructure damage is **educational infrastructure damage (**during the reported time).
- Most **displacement orders** happened **on weekdays.** , which overlap with exam and class schedules.
- War events related to food insecurity were highest in **Deir el-Balah** and **Gaza City** during the reported time (October 2023 to mid-September 2024).
- During the period **(October 2023 to mid-September 2024)** of the war, there were around 150 events that affected 'Agricultural Lands'(crops and lands), as well as 24 events that affected markets and 24 events that affected bakeries (food supply chains) - which reflects the food insecurity.
- **96.5%** of the daily casualties are civilians, with 34.7% Children, 23.9% women, and 41.4% men.

We are currently on a halt, as we do not have students' performance data to conduct further analysis and correlate our war-event findings to students' academic performance.

What we are asking for

From what we know, the use of the online learning platform (Moodle) became more frequent after the war that occurred on October 7th, and according to the website, Alaqsa-University-Gaza students are also using Moodle.

We are looking forward to getting students' data that demonstrates their interaction on Moodle to be able to measure their engagement, performance, and attendance. This includes Moodle logs capturing various student interactions, with fields such as:

- User ID (can be anonymized for data privacy).
- Timestamp.
- Event type (e.g., view, submit, login, forum post).
- Course information: title, description, students' level.
- Module.
- Session information: Duration on page, clicks, and downloads for materials and videos.
- Grade items.
- Cohort or department identifiers.
- Gender.
- Level of study (when linked via user profiles).
- Place of residence (City or address).

We would prefer students' data before and during the war so that we can compare their activity, performance, and attendance. However, in the case of the unavailability of data before the war, we are still interested in getting students' interaction data during the war.

Sample Structure:

*During the war***:**

Date: Starting from the date of the first class held during the war (the comeback of formal studying).

Number of semesters: all semesters that took place during the war.

*Before the war***:**

Date: according to the semester.

Number of semesters: the same number of semesters that took place before the war.

Targeted Faculties: Medical sciences, and Computer and Information Technology.

Number of Students: Around 1,500 students of each gender.

Number of Courses per Student: 3–4 of the most important specialization courses.

Student Distribution: From all academic levels (first, second, third, fourth—graduate). All levels are included for the Faculty of Medical Sciences.

Type of Data:

- Detailed activity logs of students on Moodle (sessions, clicks, assignment submissions, quizzes and exams, grades), and any other details relevant to the study.
- Summary and information about each course within the target sample.
- Summary of each semester: start date, end date, and general academic calendar.

Sampling Strategy:

We aim for appropriate representation from each faculty and department, taking into account the following:

- Total number of students
- Courses registered online (remotely)
- Level of activity and engagement on Moodle
- Stratification: Faculty → Department → Year of study

If data is available for 3,000 students, a sample can be drawn to reflect the distribution of students across the two faculties, taking into consideration gender (male and female), academic levels, geographic distribution (residence), current displacement location (if applicable), and academic performance (GPA or academic standing).

We are aware that the proportion of medical sciences students could be lower than that of Computer and Information Technology students.

Example:
If 85% of students are from the Faculty of Computer and Information Technology (such as: 42.5% from the Computer and Information Sciences department, 42.5% from the Engineering and Networking department), and 15% from the Faculty of Medical Sciences, then in a sample of 3,000 students:

- 1800 from the Faculty of Computer and Information Technology → distributed across its departments.(1275 from the department of Computer and Information Sciences , 1275 from department of Engineering and networking)
- 450 from the Faculty of Medical Sciences, distributed across its departments.

SUMMARY

We would appreciate data on the same 3,000 students before and during the war regarding their interaction and activity on Moodle, as stated above.